PaddyGuard AI

Prediction Metrics Explanation Guide

ප්‍රතිඵල ගණනය කරන ආකාරය පිළිබඳ මාර්ගෝපදේශය

முடிவு அளவுகள் எவ்வாறு கணக்கிடப்படுகின்றன - வழிகாட்டி

For users, reviewers and research presentations

IMPORTANT

This guide explains the recommended transparent calculation method. The application should display only values that are actually produced by the implemented and validated backend logic.

මෙම මාර්ගෝපදේශය නිර්දේශිත සහ පැහැදිලි ගණනය කිරීමේ ක්‍රමය විස්තර කරයි. යෙදුමෙන් පෙන්විය යුත්තේ backend එකේ සැබවින්ම ක්‍රියාත්මක කර validate කර ඇති ගණනය කිරීම් පමණි.

இந்த வழிகாட்டி பரிந்துரைக்கப்பட்ட வெளிப்படையான கணக்கீட்டு முறையை விளக்குகிறது. பயன்பாட்டில் காட்டப்படும் மதிப்புகள் உண்மையில் backend-ல் செயல்படுத்தப்பட்டு சரிபார்க்கப்பட்ட கணக்கீடுகளிலிருந்து மட்டுமே வர வேண்டும்.

1. What the user should understand | 1. පරිශීලකයා තේරුම් ගත යුතු දේ | 1. பயனர் புரிந்துகொள்ள வேண்டியது

English: Disease and Confidence come directly from the AI classification model. Severity, Climate Risk, Loss Risk and Estimated Loss are derived metrics and therefore must be explainable with their inputs and formulas.

සිංහල: Disease සහ Confidence යන අගයන් AI classification model එකෙන් සෘජුවම ලැබේ. Severity, Climate Risk, Loss Risk සහ Estimated Loss යන අගයන් ගණනය කර ලබාගන්නා derived metrics නිසා ඒවාට භාවිතා කළ inputs සහ formula පැහැදිලිව පෙන්විය යුතුය.

தமிழ்: Disease மற்றும் Confidence மதிப்புகள் AI classification model-இலிருந்து நேரடியாக கிடைக்கும். Severity, Climate Risk, Loss Risk மற்றும் Estimated Loss ஆகியவை கணக்கிட்டு பெறப்படும் derived metrics என்பதால் பயன்படுத்திய inputs மற்றும் formula தெளிவாக விளக்கப்பட வேண்டும்.

2. Disease classification & confidence | 2. රෝග හඳුනාගැනීම සහ විශ්වාස අගය | 2. நோய் வகைப்பாடு மற்றும் நம்பிக்கை

Example: Brown Spot - Confidence 100.00%

English: The uploaded rice-leaf image is processed by the trained classification model. The class with the highest model probability is shown as the predicted disease. Confidence is the model probability assigned to that selected class.

සිංහල: Upload කරන ලද වී පත්‍ර රූපය trained classification model එකෙන් process කරයි. වැඩිම probability එක ලැබෙන class එක predicted disease ලෙස පෙන්වයි. Confidence යනු එම තෝරාගත් class එකට model එක ලබාදුන් probability අගයයි.

தமிழ்: பதிவேற்றப்பட்ட நெல் இலைப் படம் பயிற்சி பெற்ற classification model மூலம் செயலாக்கப்படுகிறது. அதிக probability பெற்ற class நோயாக காட்டப்படும். Confidence என்பது தேர்ந்தெடுக்கப்பட்ட class-க்கு model வழங்கிய probability ஆகும்.

3. Disease severity | 3. රෝග තීව්‍රතාව (Severity) | 3. நோய் தீவிரம் (Severity)

English: Severity should represent the proportion of the leaf area that is affected by visible disease symptoms. A segmentation mask or another validated lesion-area method is preferable for this calculation.

සිංහල: Severity යනු පත්‍රයේ දෘශ්‍ය රෝග ලක්ෂණ වලින් බලපෑමට ලක්වූ ප්‍රදේශයේ ප්‍රතිශතයයි. මේ සඳහා segmentation mask එකක් හෝ validate කළ lesion-area method එකක් භාවිතා කිරීම වඩා සුදුසුය.

தமிழ்: Severity என்பது இலைப்பரப்பில் நோய் அறிகுறிகளால் பாதிக்கப்பட்ட பகுதியின்

சதவீதத்தை குறிக்க வேண்டும். இதற்காக segmentation mask அல்லது சரிபார்க்கப்பட்ட lesion-area முறை பயன்படுத்துவது சிறந்தது.

Recommended formula

Severity (%) = (Diseased / affected leaf area ÷ Total valid leaf area) × 100

Example: 3,982 affected pixels ÷ 48,500 leaf pixels × 100 ≈ 8.21%

Important validation note

English: A label such as Mild, Moderate or Severe can be assigned using thresholds that are documented and validated for the project. Example thresholds must not be presented as scientific standards unless supported by literature or expert validation.

සිංහල: Mild / Moderate / Severe වැනි label ලබාදීමට project එකේ document කර validate කළ thresholds භාවිතා කළ හැක. Literature හෝ expert validation නොමැතිව example thresholds විද්‍යාත්මක standard ලෙස පෙන්විය නොහැක.

தமிழ்: Mild / Moderate / Severe போன்ற labels வழங்க project-ல் ஆவணப்படுத்தப்பட்டு சரிபார்க்கப்பட்ட thresholds பயன்படுத்தலாம். Literature அல்லது expert validation இல்லாமல் உதாரண thresholds-ஐ அறிவியல் தரநிலையாகக் காட்டக் கூடாது.

4. Climate risk | 4. දේශගුණික අවදානම (Climate Risk) | 4. காலநிலை அபாயம் (Climate Risk)

English: Climate Risk estimates how favorable the current environmental conditions are for the predicted disease. It may use temperature, humidity, rainfall and other weather variables relevant to the disease.

සිංහල: Climate Risk යනු predicted disease එක වර්ධනය වීමට දැනට පවතින පරිසර තත්ත්ව කෙතරම් හිතකරද යන්න මැනීමකි. Temperature, humidity, rainfall සහ disease එකට අදාළ වෙනත් weather variables භාවිතා කළ හැක.

தமிழ்: Climate Risk என்பது கணிக்கப்பட்ட நோய் வளர தற்போதைய சுற்றுச்சூழல் நிலைகள் எவ்வளவு சாதகமாக உள்ளன என்பதை மதிப்பிடுகிறது. Temperature, humidity, rainfall மற்றும் நோயுடன் தொடர்புடைய பிற weather variables பயன்படுத்தப்படலாம்.

General method

Climate Risk = validated disease-specific function of temperature + humidity + rainfall + other weather factors

The exact weights / rules must be documented and supported by data, literature or agricultural expert input.

English: For transparency, the application should show the weather inputs used, for example: Temperature 28.4 °C, Humidity 86%, Rainfall 12 mm, and the final climate risk score.

සිංහල: පැහැදිලි බව සඳහා app එකෙන් භාවිතා කළ weather inputs පෙන්විය යුතුය. උදා: Temperature 28.4 °C, Humidity 86%, Rainfall 12 mm සහ final climate risk score.

தமிழ்: வெளிப்படைத்தன்மைக்காக பயன்பாட்டில் பயன்படுத்திய weather inputs காட்டப்பட வேண்டும். உதா: Temperature 28.4 °C, Humidity 86%, Rainfall 12 mm மற்றும் இறுதி climate risk score.

5. Loss risk | 5. අස්වැන්න අහිමිවීමේ අවදානම (Loss Risk) | 5. விளைச்சல் இழப்பு அபாயம் (Loss Risk)

English: Loss Risk estimates the percentage of expected yield that may be at risk. It can combine disease type, severity and climate conditions. The combination must be based on a validated model, literature-derived relationship, or expert-defined rules.

සිංහල: Loss Risk යනු අපේක්ෂිත අස්වැන්නෙන් අවදානමට ලක්විය හැකි ප්‍රතිශතය ඇස්තමේන්තු කරයි. Disease type, severity සහ climate conditions එකතු කර භාවිතා කළ හැකි නමුත් එම සම්බන්ධය validated model එකක්, literature relationship එකක් හෝ expert-defined rules මත පදනම් විය යුතුය.

தமிழ்: Loss Risk என்பது எதிர்பார்க்கப்படும் விளைச்சலில் அபாயத்தில் இருக்கக்கூடிய சதவீதத்தை மதிப்பிடுகிறது. Disease type, severity மற்றும் climate conditions இணைக்கப்படலாம்; ஆனால் அந்த தொடர்பு validated model, literature அல்லது expert-defined rules மூலம் ஆதரிக்கப்பட வேண்டும்.

Conceptual form

Loss Risk (%) = f(Disease type, Severity, Climate Risk, optional field factors)

Do not use arbitrary weights such as 50/30/20 unless they have been justified and validated.

6. Estimated yield loss | 6. ඇස්තමේන්තු කළ අස්වැන්න අහිමිවීම | 6. மதிப்பிடப்பட்ட விளைச்சல் இழப்பு

English: Estimated Loss in kilograms cannot be calculated from a leaf image alone. The system also needs an expected yield value, obtained from the farmer, farm records, or a validated yield estimate based on farm area and expected yield per unit area.

සිංහල: Leaf image එකෙන් පමණක් Estimated Loss kg වලින් ගණනය කළ නොහැක. Farmer ලබාදෙන expected yield, farm records, හෝ farm area සහ expected yield per unit area මත

පදනම් වූ validated yield estimate එකක් අවශ්‍ය වේ.

தமிழ்: ஒரு இலைப் படம் மட்டும் வைத்து Estimated Loss-ஐ kg-ல் கணக்கிட முடியாது. Farmer வழங்கும் expected yield, farm records அல்லது farm area மற்றும் unit-area yield அடிப்படையிலான validated yield estimate தேவை.

Formula

Estimated Loss (kg) = Expected Yield (kg) × Loss Risk (%) ÷ 100

Example: 1,800 kg × 17.59% = 316.62 kg

English: The interface should also show the Expected Yield used in the calculation. Without this input, a precise kilogram loss value may look misleading.

සිංහල: Calculation එකට භාවිතා කළ Expected Yield එකත් UI එකෙන් පෙන්විය යුතුය. එම input එක නොපෙන්වා exact kg loss එකක් පෙන්වීම userට නොමඟ යවන සුළු විය හැක.

தமிழ்: கணக்கீட்டில் பயன்படுத்திய Expected Yield மதிப்பும் UI-ல் காட்டப்பட வேண்டும். அந்த input இல்லாமல் துல்லியமான kg loss காட்டுவது பயனரை தவறாக வழிநடத்தலாம்.

7. Worked example | 7. සම්පූර්ණ උදාහරණයක් | 7. முழுமையான உதாரணம்

Input / Result	Example value
Predicted disease	Brown Spot
Confidence	100.00%
Affected leaf area	8.21%
Severity label	Mild (project-defined threshold)
Temperature	28.4 °C
Humidity	86%
Rainfall	12 mm
Climate Risk	59.69% (from validated climate-risk logic)
Loss Risk	17.59% (from validated yield-loss logic)
Expected Yield	1,800 kg
Estimated Loss	1,800 × 0.1759 = 316.62 kg

Example note

English: This example is for explaining the calculation flow. The exact climate-risk and loss-risk values must come from the actual validated implementation used by PaddyGuard AI.

සිංහල: මෙම example එක calculation flow එක පැහැදිලි කිරීමටයි. Climate Risk සහ Loss Risk exact values PaddyGuard AI හි සැබෑවටම implement කර validate කළ logic එකෙන් ලැබිය යුතුය.

தமிழ்: இந்த உதாரணம் calculation flow விளக்கத்திற்காக மட்டுமே. Climate Risk மற்றும் Loss Risk மதிப்புகள் PaddyGuard AI-ல் உண்மையில் செயல்படுத்தப்பட்டு சரிபார்க்கப்பட்ட logic-இலிருந்து வர வேண்டும்.

8. Recommended “More Information” text in the app | 8. App එකේ “More Information” සඳහා නිර්දේශිත text | 8. App-இல் “More Information” பரிந்துரைக்கப்பட்ட text

Full version

English: Want to know how Severity, Climate Risk, Loss Risk and Estimated Loss were calculated? Please check the PaddyGuard AI Prediction Metrics Explanation Guide for formulas, input factors and calculation details.

සිංහල: Severity, Climate Risk, Loss Risk සහ Estimated Loss ගණනය කළේ කොහොමද කියලා දැනගන්න අවශ්‍යද? Formula, භාවිතා කළ inputs සහ calculation details සඳහා PaddyGuard AI Prediction Metrics Explanation Guide එක බලන්න.

தமிழ்: Severity, Climate Risk, Loss Risk மற்றும் Estimated Loss எவ்வாறு கணக்கிடப்பட்டது என்பதை அறிய விரும்புகிறீர்களா? Formula, பயன்படுத்திய inputs மற்றும் calculation details-க்கு PaddyGuard AI Prediction Metrics Explanation Guide-ஐ பார்க்கவும்.

Short button / link text

English: More information: View Calculation Guide

සිංහල: වැඩි විස්තර: Calculation Guide බලන්න

தமிழ்: மேலும் தகவல்: Calculation Guide-ஐ பார்க்கவும்

9. What each info icon should show | 9. එක් එක් info icon එකෙන් පෙන්විය යුතු දේ | 9. ஒவ்வொரு info icon-லும் காட்ட வேண்டியது

Metric	User-facing explanation	Implementation requirement
Severity	Affected leaf area percentage calculated using a validated lesion-area method.	Validate segmentation / lesion mask.
Climate Risk	Environmental suitability score based on weather inputs relevant to the predicted disease.	Show temperature, humidity, rainfall and source/time.
Loss Risk	Estimated percentage of expected yield at risk based on validated disease, severity and climate relationships.	Show the model/rule version where practical.
Estimated Loss	Expected Yield × Loss Risk.	Always show the Expected Yield used.

10. Research / panel defence checklist | 10. Research / panel එකට defend කිරීමට checklist | 10. Research / panel பாதுகாப்பு checklist

- **Disease & confidence:** Name the trained classifier and how confidence is produced.
- **Severity:** Explain affected-area extraction method and validation.
- **Climate risk:** List exact weather variables, source, time window, rules/model and validation.
- **Loss risk:** Explain the relationship/model that converts disease conditions into yield-loss risk.
- **Estimated loss:** Show expected yield source and the exact multiplication.
- **Versioning:** Store formula/model version with the Case ID so results can be reproduced.
- **Uncertainty:** Avoid presenting estimates as guaranteed outcomes; label them as estimates/predictions.

PaddyGuard AI - Transparent and Explainable Prediction Results